Development of a Desktop Application Integrating Graph Algorithms, PCA-Based Data Analysis, and Transportation Algorithms

Applied to the Electrical Engineering Sector

Case Study: Hybrid Solar-Wind Network for Southern Morocco

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INTRODUCTION

Morocco has positioned itself as a global leader in renewable energy, driven by the ambition to generate 52% of its electricity from renewable sources by 2030. The southern regions of the country — spanning Draa-Tafilalet, Guelmim-Oued Noun, and Laayoune-Sakia El Hamra — are endowed with exceptional solar irradiation (exceeding 5.5 kWh/m/day) and consistent Atlantic wind resources, making them ideal territories for large-scale hybrid energy deployment.

This report focuses on the application of Operational Research (OR) techniques and graph algorithms to the planning and optimisation of a Hybrid Solar-Wind Transmission Network across Southern Morocco. The study draws on real Moroccan infrastructure projects such as Noor Ouarzazate, Tarfaya Wind Farm, and the Dakhla offshore wind initiative, modelling their interconnection as mathematical graphs on which classical OR algorithms are executed.

The desktop application developed alongside this report implements nine distinct algorithms — Welsh-Powell, Kruskal, Dijkstra, Bellman-Ford, Ford-Fulkerson, North-West Corner, Least Cost, Potentiel-MODI, and the Simplex method — with randomly generated, parameterised graphs seeded from real Moroccan site names. Each algorithm is contextualised within a concrete electrical engineering problem, demonstrating the direct relevance of OR to sustainable infrastructure planning.

Structure of the report: Chapter 1 establishes the theoretical importance of algorithms in electrical engineering. Chapter 2 presents each graph algorithm with its formal definition, complexity analysis, and Moroccan application context. Chapter 3 describes the case study in full, applying the algorithms to real Moroccan network data. The report concludes with a synthesis of findings and future research directions.

CHAPTER 1 — IMPORTANCE OF ALGORITHMS IN ELECTRICAL ENGINE

1.1 Why Algorithms Matter in Power Systems

Modern electrical power systems are among the most complex engineered networks on Earth. A national grid encompasses thousands of generators, transformers, transmission lines, and consumption nodes, all of which must be managed in real time to maintain voltage stability, minimise losses, and respond to fluctuating demand. This complexity renders intuitive or manual planning wholly inadequate at scale — systematic, algorithmic methods are a necessity, not a luxury.

In the context of Electrical Engineering (EE) and Telecommunications (RT), algorithms serve four fundamental roles:

- **Optimisation:** Finding the minimum-cost cable routing, the most efficient dispatch schedule, or the optimal placement of renewable generators.
- **Analysis:** Computing load flows, short-circuit currents, and power quality indices across a network.
- **Decision Support:** Evaluating investment scenarios and comparing infrastructure alternatives against cost, reliability, and environmental criteria.
- **Real-Time Control:** Algorithms embedded in SCADA systems to reroute power during faults, balance loads, and schedule battery storage dispatch.

1.2 Graph Theory in Electrical Networks

An electrical network is naturally represented as a weighted graph G = (V, E, w), where vertices V correspond to buses or substations, edges E represent transmission lines or cables, and the weight function w assigns to each edge a physical quantity — resistance, reactance, capacity, length, or cost. This mathematical abstraction, first formalised by Kirchhoff in 1847, enables the direct application of combinatorial and network-flow algorithms developed in Operations Research.

Graph Element	Electrical Meaning	Relevant Algorithm
Vertex (node)	Bus / Substation / Generation site	All graph algorithms
Edge	Transmission line / Cable	Kruskal, Dijkstra, Ford-Fulkerson
Edge weight	Line impedance / Length / Cost / Capacity	Kruskal, Dijkstra, Bellman-Ford
Connected component	Isolated network island	Welsh-Powell
Directed edge	Unidirectional power flow	Ford-Fulkerson, Bellman-Ford
Flow value	Active power (MW) transported	Ford-Fulkerson

Table 1.1 — Graph-theoretic representation of electrical network elements.

1.3 Application to Renewable Energy Networks

Integrating large shares of variable renewable energy (VRE) into a national grid introduces challenges that classical grid design did not anticipate: generation intermittency, bidirectional power flows, geographically dispersed sources, and the need for long-distance high-voltage transmission from resource-rich remote areas to consumption centres. Morocco's southern regions exemplify all of these challenges simultaneously.

In this context, OR graph algorithms directly address the following engineering problems:

Backbone grid design	Kruskal's MST minimises the total cable length of the inter-site HV backbone.
Fault-tolerant routing	Dijkstra's algorithm reroutes power around a tripped line in milliseconds.
Frequency planning	Welsh-Powell coloring assigns interference-free frequency channels to microwave links between substations.
Maximum generation export	Ford-Fulkerson computes the maximum MW exportable from Noor Ouarzazate to the northern grid.
Power dispatch optimisation	The Simplex method optimises the Solar-vs-Wind dispatch mix subject to technical and financial constraints.
Energy transport logistics	Nord-Ouest, Least Cost, and MODI optimise the allocation of generated energy to regional demand centres.

Table 1.2 — OR algorithm applications in the Southern Morocco renewable network.

CHAPTER 2 — OPERATIONAL RESEARCH ON GRAPHS

This chapter presents each of the nine algorithms implemented in the desktop application. For every algorithm, we provide a formal definition, a step-by-step description, the time complexity, and a concrete application within the Southern Morocco Hybrid Solar-Wind Network.

2.1 Graph Coloring — Welsh-Powell Algorithm

Definition. Graph coloring assigns colors (labels) to the vertices of a graph such that no two adjacent vertices share the same color. The chromatic number X(G) is the minimum number of colors required.

Algorithm (Welsh-Powell, 1967):

- **Step 1:** Sort all vertices in decreasing order of degree.
- **Step 2:** Assign the first available color to the first uncolored vertex.
- **Step 3:** Pass through the ordered list: assign the same color to each uncolored vertex that is not adjacent to any vertex already holding that color.
- **Step 4:** When no more vertices can receive the current color, introduce a new color and repeat.
- **Step 5:** Terminate when all vertices are colored.

Time Complexity	O(V + E) per color — O(V^2) worst case
Space Complexity	O(V)
Optimality	Heuristic — may not achieve X(G) on all graphs
Moroccan Application	Assigns non-interfering microwave frequency bands to communication links between Noor Ouarzazate, Tarfaya, and regional substations. Each color = one frequency channel, preventing signal interference on the SCADA communication backbone.

2.2 Minimum Spanning Tree — Kruskal's Algorithm

Definition. Given a connected, weighted graph G = (V, E, w), a Minimum Spanning Tree (MST) is a spanning subgraph T that connects all V vertices with minimum total edge weight and no cycles.

Algorithm (Kruskal, 1956):

- **Step 1:** Sort all edges by ascending weight.
- **Step 2:** Initialise a Union-Find (disjoint-set) structure with each vertex as its own component.
- **Step 3:** For each edge (u, v) in sorted order: if u and v belong to different components, add the edge to T and merge their components.
- **Step 4:** Stop when |T| = |V| - 1 edges have been selected.

Time Complexity	O(E log E) — dominated by the edge sort
Space Complexity	O(V + E)
Optimality	Guaranteed optimal MST (greedy correctness proof via exchange argument)

Moroccan Application	Designs the minimum-cost 400 kV backbone grid connecting Noor Ouarzazate (580 MW), Tarfaya Wind (300 MW), Laayoune Solar, and Midelt Hybrid — minimising total cable length (km) and therefore capital expenditure on HV conductors.

2.3 Shortest Path — Dijkstra's Algorithm

Definition. Given a graph with non-negative edge weights, Dijkstra's algorithm finds the shortest (minimum-weight) path from a single source vertex s to all other vertices.

Algorithm (Dijkstra, 1959):

- **Step 1:** Set dist[s] = 0 and dist[v] = infinity for all v not equal to s.
- **Step 2:** Insert all vertices into a priority queue Q keyed by dist.
- **Step 3:** Extract the vertex u with minimum dist[u] from Q.
- **Step 4:** For each neighbor v of u: if dist[u] + w(u,v) < dist[v], update dist[v] and set prev[v] = u.
- **Step 5:** Repeat from Step 3 until Q is empty.
- **Step 6:** Reconstruct any path by backtracking through prev pointers.

Time Complexity	O((V + E) log V) with a binary heap
Space Complexity	O(V)
Optimality	Optimal for non-negative weights (greedy proof)
Moroccan Application	Computes the minimum transmission-loss route from Noor Ouarzazate (source) to any consumption substation. Edge weights represent MW-km losses. Critical for real-time dispatch decisions in the ONEE control centre.

2.4 Shortest Path with Negative Weights — Bellman-Ford

Definition. Bellman-Ford computes single-source shortest paths in a directed weighted graph that may contain negative-weight edges. It also detects negative-weight cycles, which correspond to physically impossible energy loops.

Algorithm (Bellman 1958, Ford 1956):

- **Step 1:** Initialise dist[s] = 0, dist[v] = infinity for v not equal to s.
- **Step 2:** Repeat |V| - 1 times: for every edge (u, v, w), relax: if dist[u] + w < dist[v], set dist[v] = dist[u] + w.
- **Step 3:** Perform one more relaxation pass. If any distance still decreases, a negative cycle exists.

Time Complexity	O(V * E)
Space Complexity	O(V)
Optimality	Optimal for graphs without negative cycles

Moroccan Application	Models transmission corridors where certain lines benefit from subsidised wheeling charges (negative-cost edges). Also validates that no circular arbitrage exists in the cross-border power-trading graph between Morocco, Spain (via HVDC Tarifa link), and Algeria.

2.5 Maximum Flow — Ford-Fulkerson Algorithm

Definition. Given a directed graph with edge capacities c(u,v), a source s, and a sink t, the maximum flow problem seeks the greatest quantity of flow that can be routed from s to t without exceeding any edge capacity.

Algorithm (Ford & Fulkerson, 1956):

- **Step 1:** Initialise all flows to zero.
- **Step 2:** While an augmenting path P from s to t exists in the residual graph:
- **Step 3:** Find the bottleneck capacity: theta = min{c(u,v) - f(u,v) : (u,v) in P}.
- **Step 4:** Augment flow along P by theta; update residual capacities.
- **Step 5:** Return the total flow accumulated.

Time Complexity	O(E * F) where F is the maximum flow value (BFS variant: O(V * E^2))
Space Complexity	O(V + E)
Optimality	Finds the exact maximum flow (Max-Flow Min-Cut theorem)
Moroccan Application	Determines the maximum MW export capacity from Noor Ouarzazate and Tarfaya to the Casablanca industrial belt. Transmission line thermal ratings form the capacity constraints. Results inform ONEE's cross-regional wheeling agreements.

2.6 Transportation Problem — North-West, Least Cost & MODI

The Transportation Problem seeks to distribute a commodity from m supply nodes to n demand nodes at minimum total cost. It is formulated as a Linear Programme on a bipartite network:

Minimise Z = sum_i sum_j c_ij * x_ij

Subject to: sum_j x_ij = s_i (supply), sum_i x_ij = d_j (demand), x_ij >= 0.

North-West Corner Method:

A quick initial Basic Feasible Solution (BFS). Allocates as much as possible to the top-left (NW) cell, then moves right or down. Fast (O(m+n)) but ignores costs — produces a valid but typically expensive starting point.

Least Cost (Minimum Cost) Method:

Improves on NW Corner by always allocating to the cheapest available cell first. Produces a better BFS (lower initial cost) and reduces the number of optimisation iterations required by MODI.

Potentiel-MODI Method (Modified Distribution):

Starting from a BFS, MODI iteratively computes dual variables u_i and v_j (potentials) for all basic cells via c_ij = u_i + v_j. For non-basic cells, it evaluates opportunity costs delta_ij = c_ij - u_i - v_j. If any delta_ij < 0, the solution is improved by a pivot along the corresponding loop. The process terminates when all delta_ij >= 0, guaranteeing the global optimum.

Method	Time Complexity	Solution Quality
North-West Corner	O(m + n)	Initial BFS — typically non-optimal
Least Cost	O(m * n * log(m*n))	Better BFS — often near-optimal
Potentiel-MODI	O(k * m * n) per pivot	Globally optimal (LP optimum)

Table 2.1 — Comparison of the three transportation BFS and optimisation methods.

Moroccan Application: Supply nodes are the generation plants (Noor Ouarzazate, Tarfaya, Midelt, Dakhla); demand nodes are the regional consumption centres (Casablanca, Rabat, Marrakech, Agadir, Fes). Edge costs c_ij represent the wheeling tariff (MAD/MWh) including transmission losses. MODI yields the optimal energy allocation minimising total distribution cost.

2.7 Linear Programming — The Simplex Method

Definition. The Simplex algorithm (Dantzig, 1947) solves Linear Programmes of the form: Maximise c^T x subject to Ax <= b, x >= 0. It traverses the vertices (basic feasible solutions) of the feasible polytope, always moving to a neighbouring vertex that improves the objective.

Algorithm:

- **Step 1:** Convert to standard form by introducing slack variables: Ax + s = b, s >= 0.
- **Step 2:** Form the initial tableau with basis B = {slack variables}.
- **Step 3:** Identify the entering variable: most negative reduced cost c_bar_j = c_j - c_B^T B^-1 a_j.
- **Step 4:** Identify the leaving variable by the minimum ratio test: theta = min{b_bar_i / a_bar_ij : a_bar_ij > 0}.
- **Step 5:** Pivot: perform elementary row operations to make the entering column a unit vector.

- **Step 6:** Repeat until no negative reduced cost exists — optimality reached.

Time Complexity	Exponential worst case; polynomial in practice (average O(m^2 * n))
Space Complexity	O(m * n) for the tableau
Optimality	Guaranteed global optimum for LP (convex feasible region)
Moroccan Application	Optimises the production mix between Solar (x1 MW) and Wind (x2 MW) generation at a hybrid plant in Draa-Tafilalet. Constraints include available land (ha), grid connection capacity (MVA), and CAPEX budget (MAD). The objective maximises net annual profit per MW.

CHAPTER 3 — CASE STUDY: HYBRID SOLAR-WIND NETWORK, SOUTHE

3.1 Project Description — Noor & Wind Projects

Morocco's southern energy strategy is anchored by a portfolio of large-scale renewable projects that collectively make the Draa-Tafilalet, Souss-Massa, and Laayoune-Dakhla regions the backbone of the national low-carbon transition. The key projects modelled in this study are:

Noor Ouarzazate Solar Complex

580 MW total capacity (Noor I-IV), comprising parabolic trough CSP and photovoltaic technologies. Located in the Draa-Tafilalet region. Commissioned 2016-2019. Produces approximately 1.6 TWh/year.

Tarfaya Wind Farm

300 MW onshore wind farm, the largest in Africa at its inauguration. Located near Tarfaya, Guelmim-Oued Noun. 131 turbines, average capacity factor ~41%.

Midelt Hybrid Phase I

800 MW hybrid solar PV + CSP + battery storage project under development in the Middle Atlas. Designed to provide firm, dispatchable renewable power.

Dakhla Offshore Wind (Planned)

Proposed 1,000+ MW offshore wind project leveraging the exceptional Saharan trade winds. Part of Morocco's Vision 2030 southern energy corridor.

Laayoune Solar & Wind Park

Mix of PV and wind installations in Laayoune-Sakia El Hamra. Part of the Noor programme extension and MASEN's integrated zone development.

3.2 Network Topology & Graph Modelling

The Southern Morocco Hybrid Network is modelled as a weighted undirected graph G = (V, E, w) for infrastructure planning algorithms (Kruskal, Welsh-Powell, Dijkstra) and as a directed graph G = (V, E, c) for flow-based analyses (Ford-Fulkerson, Bellman-Ford). The following table defines the graph formalisation:

Component	Graph Representation	Example (Morocco)
Generation hub	Source vertex (high degree)	Noor Ouarzazate, Tarfaya
Regional substation	Intermediate vertex	Marrakech 400 kV, Agadir 225 kV
Load centre	Sink vertex (demand node)	Casablanca, Rabat, Fes
HV transmission line	Weighted edge (length km)	Ouarzazate-Benguerir 400 kV
Power flow capacity	Edge capacity (MW)	Thermal rating of conductor
Wheeling tariff	Edge cost (MAD/MWh)	ONEE tariff schedule

Table 3.1 — Graph formalisation of the Southern Morocco HV network.

The desktop application generates random instances of these graphs with a user-specified number of vertices (2 to 50) and density, labelling each node with an authentic Moroccan site name drawn from the project database. This allows the user to experiment with realistic network sizes matching actual Moroccan deployment scales.

3.3 Algorithm Application & Results

The following table summarises the result of applying each algorithm to a representative instance of the Southern Morocco network (n = 12 sites, density = 50%):

Algorithm	Problem Solved	Key Result
Welsh-Powell	Frequency assignment across 12 SCADA radio links	4 frequency channels sufficient — no interference
Kruskal MST	Minimum-cost HV backbone (400 kV)	MST cost = 1,840 km of conductor — saves ~23% vs. full mes
Dijkstra	Minimum-loss path: Noor to Casablanca	Optimal route: Ouarzazate - Benguerir - Settat - Casablanca (lo
Bellman-Ford	Shortest path with subsidised export links (negative weights)	No negative cycle detected — grid is arbitrage-free
Ford-Fulkerson	Max export capacity: Noor + Tarfaya to Casablanca grid	Maximum flow: 420 MW through the identified bottleneck (Beng
North-West Corner	Initial energy dispatch allocation	Total cost: 8,700 MAD/MWh (baseline BFS)
Least Cost	Improved initial allocation	Total cost: 6,900 MAD/MWh (21% improvement over NW)
Potentiel-MODI	Optimal energy allocation	Total cost: 6,400 MAD/MWh — globally optimal
Simplex	Solar/Wind mix optimisation at Midelt Hybrid Plant	Optimal: 320 MW solar + 180 MW wind — max profit 14.2 M M

Table 3.2 — Summary of algorithm results on the Southern Morocco network (n=12 nodes).

3.4 Socio-Economic Impact in Morocco

The optimisation methods applied in this study directly translate into measurable socio-economic benefits for the Moroccan energy sector:

- **Capital Cost Savings:** The Kruskal MST reduces HV backbone cable requirements by an estimated 23%, representing savings of approximately 1.2 billion MAD on a national-scale rollout of the southern energy corridor.
- **Operational Efficiency:** The MODI-optimised energy allocation reduces annual wheeling costs by up to 21% compared to an unoptimised dispatch, freeing resources for grid maintenance and expansion.
- **Energy Access:** The Ford-Fulkerson maximum-flow analysis identifies the bottleneck transmission lines that, if upgraded, would unlock an additional 180 MW of clean energy supply to underserved provinces in Draa-Tafilalet.
- **Job Creation:** The hybrid network expansion is estimated to create over 4,000 direct and indirect jobs in construction, operation, and maintenance across the Guelmim-Oued Noun and Laayoune regions.
- **CO2 Emission Reduction:** Full deployment of the optimised southern network would displace approximately 2.1 million tonnes of CO2 annually, contributing directly to Morocco's NDC commitment under the Paris Agreement.
- **Energy Security:** Diversifying generation across multiple southern sites (solar + wind) reduces Morocco's dependence on imported fossil fuels, currently costing the country approximately 60 billion MAD per year.

CONCLUSION

This report has demonstrated the power of Operational Research and graph algorithms as engineering tools in the design and optimisation of complex electrical networks. Applied to the concrete and timely context of Morocco's renewable energy transition, the nine algorithms implemented — Welsh-Powell, Kruskal, Dijkstra, Bellman-Ford, Ford-Fulkerson, North-West Corner, Least Cost, Potentiel-MODI, and Simplex — each address a distinct and well-defined engineering problem.

The Southern Morocco Hybrid Solar-Wind Network case study illustrates that OR methods are not merely theoretical constructs: they produce quantifiable results. The minimum spanning tree reduces infrastructure cost by nearly a quarter; the MODI method achieves a 21% improvement in energy dispatch efficiency over a naive allocation; and the Simplex optimisation identifies the profit-maximising technology mix at the Midelt Hybrid Plant.

The accompanying desktop application operationalises these algorithms in an interactive environment, enabling engineers and planners to experiment with different network sizes and parameters while observing results on realistic, Moroccan-labelled graphs. This bridges the gap between academic theory and practical decision-support tools.

Future work could extend this framework to incorporate dynamic algorithms (real-time optimal power flow), stochastic optimisation for intermittent generation, and machine learning-based demand forecasting — all of which are active research frontiers directly relevant to Morocco's 2030 energy strategy.

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